Pre-A Route Decision, Controller-Free CL8 Macro, Routed Seam, and State Split

TECT verification-first repository
claim `C6-SPACETIME-SIGNATURE`

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1 Purpose and scope

This note answers two linked questions.

First, should the A13/T-050 controlled-shell programme remain the dominant research route before a physical functional has been selected? Second, what does the controller-free two-kick CL8 construction actually prove about open characteristic cuts, periodic seams, and invariant states?

The answer is deliberately asymmetric. The existing A13 mathematics is preserved, including the exact R-166 v1.1 sparse-fibre margin, but T-050 is parked from the main physical priority. The primary route becomes the T-054 Pre-A evidence and candidate-admission programme. In parallel, the CL8 branch advances substantially: the controller is removed, all open acyclic monotone cuts are reconstructed, and an exact all-k swap-dressed routed seam conjugacy is proved. An exact C4 witness refutes a universal positive quadratic invariant, tangent-circuit Gaussian, or zero-centred nonlinear Gaussian conclusion. It does not exclude arbitrary off-centre nonlinear stationary Gaussians.

All results in this note are claim-nonbearing T0 results. They do not select a physical equation, a BCC phase, a vacuum, a causal structure, or a preferred state.

2 Dependency and priority audit

2.1 What is already closed

The claim A5-SECTOR-A-SYNTHESIS is a published T6 conditional-composition theorem under exactly seven named hypotheses. Its empty open-gate list means that theorem is closed in its declared scope. It does not prove that the current production functional is nature's canonical law, and it does not contain the full derivative Class-II interacting measure.

The A6–A13 cards form a separate SA-F4 constructive-measure programme. In the currently registered route, A13/T-050 lies before the source-overlap, Nelson exponential, removal, and interacting-measure gates. This makes T-050 necessary for that route, not logically necessary for every possible constructive theory and not a dependency of A5.

R-166 v1.1 is retained exactly in its proved scope:

$$K_{\text{owner}}(g) > -\frac{4}{5}I, \qquad \rho = \frac{3}{20}, \qquad m_{\text{radial}} = \frac{31}{220}, \qquad \epsilon_v = \frac{367}{880}.$$

It applies only to the fixed side-16, nonaliased R-153 sparse principal fibre with the declared regulator and floor. It is not a complete production-owner theorem.

2.2 Priority verdict

Deepening one candidate-specific constructive estimate cannot decide whether that candidate is the correct physical functional. The complete finite production cylinder required for the next A13 step is itself not yet selected by a common state, reference, observable, and counterterm contract. Continuing to add abstract owner lemmas now would risk solving the hardest part of an unadmitted candidate.

Therefore:

1. preserve every exact A13 and R-166 result;
2. park T-050 from the main physical priority;
3. make T-054 Pre-A the primary research path;
4. reopen T-050 as the main path only if Pre-A selects the current A1/A7 branch and freezes its complete finite production cylinder, governance explicitly restores that programme, or a scheme-independent exact or outward-certified legal counterdirection appears.

The first reopened A13 test must use one minimal legal complete finite production cylinder and count every forward, legal reverse, balanced, low, current, trace, future, forest, projected-force, and joint-sextic occurrence exactly once.

3 Round-one evidence and candidate admission

The current evidence register is a partial intake, not a charter-complete freeze. It records primary-source discovery constraints for Lorentz anisotropy, gravitational/electromagnetic propagation, superconducting flux, helium critical scaling, and energy-dependent photon propagation. Examples include https://doi.org/10.1038/ncomms9174, https://doi.org/10.3847/2041-8213/aa920c, https://doi.org/10.1038/nature08574, and https://doi.org/10.1103/PhysRevB.68.174518.

The helium exponent used for later validation is public. It is therefore a declared non-fitting validation target, not a sealed or unopened holdout.

Entry	Current role	What is established	Why selection is not licensed
M0	Effective baseline	Tested domain-specific low-energy descriptions	No single microscopic state, regulator, or common energy reference
M1	Current TECT benchmark	Scoped variational and constructive mathematics; pinned neutral nonzero mechanism fails	Common observable and physical state/reference maps are incomplete
M2	Finite-wave-number alternative	Static candidate window and inserted tree-level cone	Inertial law and physical map are inserted; quantum and reference contracts remain open
M5	Nonlocal shell alternative	Conditional static shell minimum	Bare joint T053 survivor fails through the rank-one spatial Hessian
CP1	Bridge infrastructure	Characteristic, state-transport, routed, fixed-lattice 3D comparator, and scoped phase lemmas	It is not a round-one contestant; effective reduction, physical reference, real-time ground/KMS, and continuum contracts remain incomplete

The corrected round-one result is:

PRE-A-ROUND1-PARTIAL-EVIDENCE-INTAKE-PINNED-M1-BARE-M5-SCOPED-FAILURES-AND-CURRENT-NONSELECTION.

No winner, shortlist, survivor, or Pre-A exit is authorized. The primary next gate is

PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-FREEZE,

followed by a preregistered quadratic-causal admission triage whose only outputs are PASS, FAIL, and NOT ADMITTED. Raw cross-theory energy magnitudes may not be ranked without matched volume, normalization, boundary, regulator, and reference conventions.

4 Controller-free two-kick macro theorem

Let

$$w = \frac{a}{8}, \qquad \mu = \chi w, \qquad \kappa = \frac{wc}{a^2} = \frac{c}{8a},$$

and let the bond Hamiltonian be

$$h_b = \frac{\|p\|^2 + \|r\|^2}{4\mu} + V(x,y), \qquad D_x D_y V = -\kappa I_8.$$

The exact bond-flow mass is $m = 2\mu$. The full-node drift is $D_t(q,p) = (q + tp/\mu, p)$. With

$$h = \frac{\Delta}{8}, \qquad s = \frac{\Delta}{2},$$

define

$$M_\Delta = D_h K_s D_{2h} K_s D_h. \tag{4.1}$$

This is two Strang substeps for h_b. It is controller free, polynomial, globally symplectic, and temporally invertible. Put

$$\rho = \frac{2hs\kappa}{\mu} = \frac{c\Delta^2}{8\chi a^2}. \tag{4.2}$$

Theorem 1 (Global mixed macro)**.** *If $\rho \neq 0$ and $\rho^2 \neq 1$, all four complete-leg mixed projections of M_Δ have explicit global inverses. In eight species,*

$$\det\left(\frac{\partial E}{\partial S}\Big|_W\right) = \rho^{16}, \qquad \det\left(\frac{\partial N}{\partial S}\Big|_W\right) = (1-\rho^2)^8, \tag{4.3}$$

with the two leg-exchanged counterparts. No commutation of onsite Hessians is assumed.

The proof solves the middle positions linearly before evaluating the nonlinear onsite force. The ordered Schur cancellation is $DA - \beta C = (1-\rho^2)I$, not a reordered product.

Two kicks give $2s = \Delta$, and four endpoint drift parameters give $4h = \Delta/2$. Thus every bond potential and every half-owned endpoint kinetic coefficient occurs exactly once.

This is an alternative two-kick bond-macro and routed-seam theorem. It does not reopen or replace the classical inserted-one-dimensional D–K–D common-parent result already closed by EXP-000758/759. The new macro's common regular state, physical reference, counterterm, effective-reduction, and continuum contracts remain separate open questions.

4.1 Exact-flow firewall

For the exact bond flow,

$$\det D_{(y,r)}(X,P) = \left(\frac{\kappa^2}{48\mu^2}\right)^8 t^{32} + O_K(t^{34}). \tag{4.4}$$

For the macro, the species-leading coefficient is instead

$$\frac{\kappa^2\Delta^4}{64\mu^2}. \tag{4.5}$$

The denominators 48 and 64 are intentionally different. Equation (4.1) is a second-order splitting companion, not the exact autonomous flow.

5 Open cuts and periodic seam

5.1 Open all-cut theorem

Placing the same macro at every vertex of an open directed rectangle gives a unique global assignment from west-south data. Every open acyclic monotone order-ideal cut contains the same number of complete legs and is related to the incoming arms by a global polynomial symplectic diffeomorphism. Complementary arms reconstruct each other exactly.

5.2 Raw periodic no-go

Quotienting by $T = (m, -n)$ gives a block transfer with occurrence graph $K_{n,m}$, mn gates, and row/column degrees m, n. A raw fixed-site even/odd ring period has graph C_{m+n}, $m+n$ gates, and degree two. Direct ledger equality forces $m = n = 2$. Even there, an exact rational tangent separates direct equality, although an exact C4 cut conjugacy survives.

5.3 All-k swap-dressed routed theorem

For a square k-by-k quotient, let

$$G_{j,i} = M_{\Delta;(H_j,V_i)}, \qquad [x]_k = 1 + ((x-1) \bmod k).$$

Let U_k be the square word ordered by increasing $i+j$, C_k its triangular subword $i + j \leq k$, and

$$L_\ell = \prod_{j=1}^{k} G_{j,[\ell-j]_k}, \qquad F_k = L_k \cdots L_1.$$

Every pair of noncommuting gates shares a horizontal or vertical wire. Projection onto each such dependent pair gives the same word on both sides, so the Cartier–Foata trace-monoid criterion proves

$$C_k U_k = F_k C_k. \tag{5.1}$$

Straight routing outputs (E, N), while the geometric frontier orders (N, E). Hence the geometric gate is $P \circ M_\Delta$. With slot permutations

$$\begin{aligned} \mathrm{pos}_r(H_j) &= 2(k-j) + r \pmod{2k}, \\ \mathrm{pos}_r(V_i) &= 2(i-1) + 1 - r \pmod{2k}, \end{aligned}$$

one obtains

$$C_k^{\text{ring}} U_k (C_k^{\text{ring}})^{-1} = R^{-k} \widehat{F}_k. \tag{5.2}$$

This is an exact all-k swap-dressed routed seam-frame conjugacy. It is not raw-M EO equality. The raw shortcut first fails at $k = 3$.

6 Invariant-state route split

6.1 Positive singular states

The all-zero phase is fixed. For $r < 0$, with $v = \sqrt{-r/g}$, the globally collective all-$+v$ and all-$-v$ phases are fixed. Therefore

$$\delta_0, \qquad \frac{1}{2}(\delta_{+v} + \delta_{-v}) \tag{6.1}$$

are exact positive normalized stationary classical probabilities. They are singular, nonselecting, and determine no energy reference or counterterm. Liouville volume is invariant but has infinite mass.

6.2 Conditional one-bond shadow

For one quadratic bond $V(q) = q^T K q/2$, $K > 0$, if

$$\sigma\left(\frac{hs}{\mu} K\right) \subset (0, 2),$$

then the macro preserves

$$\widetilde{H}_b(q, p) = \frac{\|p\|^2}{4\mu} + \frac{1}{2} q^T K \left[I - \frac{hs}{2\mu} K\right]^{-1} q. \tag{6.2}$$

This supplies a normalized classical bond Gibbs law. It is bond-correlated and pairing-dependent, so it does not automatically tile the periodic/open circuit.

6.3 Scoped Gaussian obstruction

For one species use $m = (q_H, p_H, q_V, p_V)$. Both implementations derive from the macro the exact tangent

$$L = \begin{pmatrix} 1/4 & 11/16 & 3/4 & 5/16 \\ -1 & 1/4 & 1 & 3/4 \\ 3/4 & 5/16 & 1/4 & 11/16 \\ 1 & 3/4 & -1 & 1/4 \end{pmatrix}. \tag{6.3}$$

The full zero-phase 32-dimensional bond tangent is verified to be eight decoupled identical species blocks. For C4 use the one-species order

$$(q_{H_1}, p_{H_1}, q_{H_2}, p_{H_2}, q_{V_1}, p_{V_1}, q_{V_2}, p_{V_2}),$$

with $F = (DA)(BC)$ acting right-to-left. At $\rho = 1/2$,

$$\det(\lambda I - F) = (\lambda - 1)^2(\lambda^2 + \lambda + 1)(\lambda^2 + 3\lambda + 1)^2. \tag{6.4}$$

The reciprocal real roots $(-3 \pm \sqrt{5})/2$ include a modulus greater than one. If $G > 0$ satisfied $F^T G F = G$, an eigenvector would force $|\lambda| = 1$, a contradiction. Hence the tangent circuit has no positive-definite quadratic invariant and no nondegenerate invariant Gaussian. For the full nonlinear fixture there is no C^2 invariant with a strict nondegenerate minimum at zero, hence no zero-centred invariant Gaussian. An arbitrary off-centre nonlinear stationary Gaussian is not excluded. The exact C4 conjugacy transfers only these scoped obstructions to the routed block.

This is `NG-2026-08-09-PRE-A-CP1-CL8-UNIVERSAL-PERIODIC-QUADRATIC- SHADOW-GIBBS`. Thus mixed invertibility is not a state-stability condition, and the common regular-state gate remains open.

The new delta measures are singular, while the one-bond shadow Gibbs law is pairing-dependent. Neither matches the inherited regular oscillator-number vacuum, Gibbs law, or reference. The existing passive-two-arm quartic-reuse, original-H-state, and principal-Floquet-Gibbs negative results therefore remain in force; no common reference match is proved here.

7 What remains for Pre-A and Sector A

The main path and the parallel branch should now be kept distinct.

Lane	Next decisive work	Completion meaning
Round-one Pre-A	Freeze evidence roles and minimum manifests; run common quadratic-causal admission triage	A defensible PASS/FAIL/NOT-ADMITTED matrix, not yet a winner
CP1 state	Bloch/Floquet blocks, positive Lyapunov equation, explicit zero-mode convention, cut covariance	A regular common finite-regulator state may become admissible
Dimensional bridge	Reuse ST8 block factorization, Q3LOCK, CL8 transverse-zero restriction, and the new macro in one commutative diagram	A legitimate 1D-to-3D parent map or an exact obstruction
Candidate-specific Sector A	Freeze the selected candidate action, state, reference, regulator, and counterterms	Only then run that candidate's A1/A2/A3 and constructive programme
SA-F4/A13	Reopen the complete finite production owner if its branch is selected or an exact scheme-independent counterdirection appears	T-050 may again become main-path work

It is misleading to give one percentage for "Sector A remaining." The published conditional A5 theorem is complete in its scope. The physical canonical-functional problem is not. The remaining work is best estimated as five major gates: evidence/admission freeze, candidate triage, common state/reference/counterterm contract, 1D-to-3D lift, and candidate-specific constructive completion. T-050 is only in the last gate if the current A1/A7 branch survives.

8 Devil's-advocate

1. **Objection: A13 was abandoned. DISMISSED.** Its exact results and current SA-F4 dependency chain are preserved. Only its physical research priority changed.

2. **Objection: A5 closure proves the physical theory. UPHELD as a prohibited inference.** A5 is a seven-hypothesis conditional composition and does not select a canonical functional.

3. **Objection: the macro determinant used the wrong mass. DISMISSED after correction.** Exact flow uses $m = 2\mu$; the full-node drift uses μ. The 48/64 firewall is explicit.

4. **Objection: the earlier one-kick rank-eight no-go was erased. DISMISSED.** One q-only kick remains rank eight. Two kicks separated by drift use the macro route that the no-go explicitly left open.

5. **Objection: open cuts equal raw periodic EO. DISMISSED.** Incidence and an exact rational witness reject direct equality. The positive theorem is swap dressed and includes the seam frame.

6. **Objection: symplecticity supplies a universal Gaussian state. DISMISSED in the proved scope.** The C4 tangent is hyperbolic, excluding a tangent Gaussian and a zero-centred nonlinear invariant Gaussian. Arbitrary off-centre nonlinear stationary Gaussians remain open.

7. **Objection: the public validation value is a sealed holdout. DISMISSED by correction.** It is recorded only as a visible non-fitting validation target.

8. **Objection: CP1 is already a candidate winner. UPHELD as false.** CP1 is currently bridge infrastructure and is not a round-one contestant.

9 Reproduction

Primary exact audit:

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_cl8_controller_free_two_kick_macro_bridge.py
--profile f0
```

Independent non-importing Fraction/jet audit:

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_cl8_controller_free_two_kick_macro_bridge_independent.py
--profile f0
```

The stored f0 results pass 171/171 and 108/108. The adverse f1 profile also passes the same counts and is stored separately. The integrated verifier reruns both f0 implementations, compares invariants and hashes, checks the negative and exploration records, and verifies the generated proof map.

10 Result footer

```
Result ID:               PA-CP1-CL8-EXACT-GLOBAL-SIDEWAYS-MACRO-AND-
                         FIXED-REGULATOR-SPLITTING-BRIDGE.
Precise statement:       Global mixed macro/open cuts; exact all-k routed
                         seam; singular states and one-bond shadow; raw-EO
                         plus scoped C4 quadratic/Gaussian routes fail.
Scope:                   T0 inserted 1D fixed regulator; claim-nonbearing.
Dependencies:            EXP-000783--EXP-000788; A5/A7/A13; CP1, Goursat,
                         ST8, and Q3LOCK authorities.
Evidence grade:          EXACT ANALYTIC + EXECUTED + INDEPENDENT + PDF QA.
Reproduction command:    python -X utf8 codes/foundations/
                         pre_a_cp1_cl8_controller_free_two_kick_macro_
                         bridge_verify.py
Expected output:         integrated 56/56; children 171/171 and 108/108;
                         aggregate 335/335; release checks PASS.
Falsification gate:      Any failed mixed inverse, all-k routed table,
                         macro-derived C4 tangent/decoupling/polynomial,
                         one-bond shadow, hash, or scope gate.
Tier before / after:     C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:  Off-centre Gaussians, autonomous flow, common
                         state/reference, 1D-to-3D, continuum, C6, CP1,
                         Pre-A, A13, and physical Sector-A remain open.
Next required action:    PA-ROUND1-EVIDENCE-ROLE-AND-MINIMUM-MANIFEST-
                         FREEZE; in parallel run CP1 Bloch/Lyapunov state
                         feasibility and the ST8/Q3LOCK dimensional lift.
```